

YASH LALA

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EDUCATION

Yale Ph.D. in Computer Science

2023 - Present

UCLA B.S. in Computer Science

GPA: 3.782, 2018 - 2022

PROFESSIONAL EXPERIENCE

NOVA Lab, Yale CS Department

Aug 2023 - Present

PhD Student

Advisor: Anurag Khandelwal

- Researching techniques for accelerating OS memory migration. Project aims to facilitate high-throughput RDMA-based far memory runtimes. Developed extensive knowledge of the the Linux kernel's memory management, RDMA, page placement, IPI, and swap subsystems. Particular focus on optimizing high throughput, low-latency runtimes, such as memory migration systems and RDMA SmartNIC network stacks. Publication to appear at **SOSP '25**.

NVIDIA

May 2025 - Aug 2025

Systems Software Intern

Supervisor: Pasha Shamis

- Developed network performance models for Mixture-of-Experts (MoE) communication phases in ML models.
- Conducted performance analysis of existing MoE library implementations (such as NCCL, DeepEP, etc), then optimized the libraries for NVIDIA's GB200 cluster hardware.
- Gained experience with LLM workload network traffic patterns, GPU performance debugging (particularly with regards to GPU-initiated RDMA network communication).

SOLAR Lab, UCLA CS Department

Sept 2021 - Sept 2022

Student Researcher

Supervisor: Harry Xu

- Volunteered during the school year, employed full-time to work on kernel patches over the summer. Developed kernel mechanisms for for transparent memory disaggregation. Worked heavily with kernel programming and debugging tools, such as QEMU + GDB, serial port debugging, and perf.
- Independently developed patches for the Linux kernel's swap subsystems, with the goal of merging these changes upstream. Patchset extends the cpuset controller to allow per-cgroup control of active swap devices. Associated refactoring has positive implications for swap throughput, and makes it easy to manage frontswap-based remote memory systems. Patchset available at github.com/yashlala/canvas-linux.
- Developed a patchset to improve the Linux kernel's physical page allocation latency. The patch reduces tail latencies by refilling the percpu low-order free page lists asynchronously using RCU.

CSSI Program, UCLA CS Department

July 2022

Tutor Undergrad (TA)

- Taught introductory data science to high school students. Led discussion sections, prepared assignments, graded papers, and advised students.

Veritas Technologies LLC

June 2021 - Sept 2021

SDE Intern

- Worked on NetBackup Flex, a platform for large-scale data consolidation and backup. Implemented automatic node discovery and cluster assimilation over datacenter networks. Primarily worked with Ansible, Docker, and various glue languages.

PUBLICATIONS

Scalable Far Memory: Balancing Faults and Evictions

August 2023 - August 2025

- Co-first author. Paper to appear at **SOSP 2025**.

GRU4RecBE: Session Based Recommendations with Features

March 2021 - June 2021

- Developed session-based recommendation system in *PyTorch* which extends the GRU4REC architecture with rich item features extracted from the pre-trained BERT architecture. Non-attentive model outperforms state-of-the-art session-based models over the MovieLens benchmark datasets. [Paper accepted to AAAI Student track](#).

PROJECTS

NDN Multicast

Feb 2022 - June 2022

- Extended routing protocols for NDN (Named Data Network) under Lixia Zhang. Extended NLSR (a link-state routing algorithm for NDN) to allow for efficient multicast delivery of NDN Interest packets. Student paper available at yashlala.com/nlsr-poster.pdf.

MISCELLANEOUS

Enjoys teaching: led student [seminar series](#), volunteered with [ACM TeachLA](#) and as a docent for the [Computer History Museum](#). Certified as a operator for the punched-card based [IBM 1401](#).